Molecular Biology and Medicine

Science

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Molecular Biology and Medicine (A13837)

Short Title: Molecular Biology and Medicine
Faculty Science and Computing
Department Science
Cluster Biology
Author ROMAHONY
Credits 5 **Level:** Advanced

Description of Module / Aims

This module describes the impact of molecular biology on our understanding of human genetics and disease. The impact of recombinant DNA technology on the diagnosis and treatment of human diseases is comprehensively covered.

Programmes

MOLB-0002	BSc (Hons) in Applied Biology with Quality Management (WD_SQMBI_B)	4 / 2 / M
MOLB-0002	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	4 / 8 / M

Indicative Content

- Introduction to human genetic disease; sex determination, linkage and mapping, chromosome mutations
- Structure & mapping human genome, genetic disorders diagnosis, screening and treatment
- Personalised medicine, theranostics & microRNAs
- Cancer as a genetic disease: molecular and immuno-based therapies
- Modern cytogenetics
- DNA genetic testing
- The role of genetics in animal testing and tailored medicine
- The role of transgenics in disease research/treatment
- Modern molecular and cellular therapies; biomolecules & vaccines
- Human genetics and society: ethical, legal and social issues

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Establish the type of defects that lead to a variety of human genetic diseases.
2. Critique the structure of the human genome and distinguish between the methods for mapping the HG.
3. Evaluate the impact of genomics and proteomics on our understanding and treatment of human disease.
4. Compare and contrast new biomarkers, biomolecules, vaccines and bio-therapeutics and justify their use/application in today's society.
5. Select and distinguish between the novel strategies of DNA-based therapies in the treatment of disease.
6. Formulate, recognise and consider the ethical, legal and social issues associated with molecular medicine.

Learning and Teaching Methods

- Lectures are delivered using a flipped classroom approach using the blended learning model.
- Students complete online e-activities prior to and post face-to-face lecture sessions.
- Formative activities include online discussion forums, audio and video files, creation of mindmaps and concept maps.
- Moodle is the platform through which such activities are delivered.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	100%	1,2,3,4,5,6

Assessment Criteria

<40%: No understanding of basic module content. Unable to carry out critical thinking and analysis.
40%-49%: Able to understand basic module concepts. Has difficulty with critical thinking and problem solving.
50%-59%: A good understanding of module content with demonstration of some critical thinking and problem solving skills.
60%-69%: A good understanding of module content and good critical thinking and problem solving skills.
70%-100%: Excellent in all areas.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Tutorial	12	
Independent Learning	99	

Supplementary Material

- Cox, T. and J. Sinclair. _Molecular Biology in Medicine_. 2nd ed. UK: Blackwell Science, 1998.
- Klug, W., M. Cummins, M. Palladino and C. Spencer. _Concepts of Genetics_. 10th ed revised. Uk: Pearson, 2012.
- Speicher , M., S. Antonarakas and A. Motulsky. _Human Genetics - Vogel and Moluskys - Problems and approaches_. Heidleberg: Springer, 2010.
- Strachan, T. and A. Read. _Human Molecular Genetics_. 4th ed. New York: Garland Science Taylor and Francis Group LLC, 2011.

Requested Resources

- Lecture Room: Loose Seated